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# Minimal required boundary conditions for the thermal spallation process of granitic rocks

Michael A. Kant, Philipp Rudolf von Rohr *

*ETH Zurich, Institute of Process Engineering, Sonneggstrasse 3, 8092 Zurich, Switzerland*

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ABSTRACT

The spallation process is based on the effect, that hard rocks with a high quartz content disintegrate into small disc-like fragments, if the rock surface is rapidly exposed to high thermal loads. Spallation is pursued as a contact-free drilling technology for various applications. In view of increasing the knowledge about the process and for determining the limitations of the applicable operating range, a profound knowledge of the minimal required boundary conditions are of significant importance. These conditions are characterized by the lowest surface temperature and heat transfer coefficient at which spallation can be successfully initiated. In order to determine the minimal required boundary conditions, spallation experiments were conducted in which granitic rock samples were rapidly heated by a methane–air burner. A novel measuring concept is proposed to measure the surface temperature, using high-speed pyrometers to temporally resolve the detachment of single spalls. The heat transfer coefficient of the impinging flame was determined by measuring the heat flux in the stagnation point of the jet using an industrial heat flux sensor. The reported data of the boundary conditions show good accordance with data published by other researches and supports their proposed characteristic for the specific heating process.

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## 1. Introduction

If certain rock types are rapidly exposed to a hot fluid jet, the upper layer of the rock will locally disintegrate into small disc-like fragments. This so-called spallation effect can be used as a contact-free drilling technology for various applications.[1–3] Additionally, spalling of rocks and concrete came recently into the focus of research on fire in tunnels.[4–6] During the spallation process, a high heat flux is transferred from the impinging hot fluid jet to the rock. Thereby, due to the low thermal conductivity of the rock, high surface temperatures are induced, leading to high thermal stresses in the upper layer of the rock. Moreover, the cold surrounding rock acts as a confinement and produces compressive stresses on the local failure zone. Due to the thermally induced stress field, the rock will locally disintegrate at naturally present flaws into small fragments called spalls (see Fig. 1).[3,7] The created spall buckles when losing the contact in the failure plane. Finally, as a result of tensile stresses at the outside of the spall, the hot spall is ejected from the surface and the spallation process continues on with the newly exposed rock surface. Thereby, the spallability will vary considerably within different rock types: hard rocks are easier to spall than soft, sedimentary rocks.[8–11] The spallation effect has been investigated by several researchers.[1–3,7,12–14] Nevertheless, the lowest boundary conditions where spallation can be initiated were so far not systematically analyzed.

### 1.1. Boundary conditions for spallation

Two main parameters for the required boundary conditions for the spallation process are reported so far: the heat flux $q$ through the rock respectively the heat transfer coefficient $h_{fl}$ between rock surface and impinging fluid jet and the induced spallation temperature $\vartheta_{SP}$ of the rock [11-14]. The spallation temperature $\vartheta_{SP}$ is defined as the surface temperature directly before the spall detaches from the surface. Right after the spall is ejected from the surface, the temperature in the failure plane of the spall $\vartheta_{FP}$ can be measured. The occurring temperature difference $\Delta\vartheta = \vartheta_{SP} - \vartheta_{FP}$ between these two temperatures is an important parameter for the investigation of the spallation process.

In several publications, measurements of the boundary conditions in spallation experiments are reported.[13,14] These reports are based on the determination of the spallation temperature $\vartheta_{SP}$ and the heat flux $q$ through the rock. In order to improve these measurements, Wilkinson [14] and other researchers proposed to use high-speed pyrometers for future measurements, which are

* Corresponding author.
*E-mail addresses:* kant@ipe.mavt.ethz.ch (M.A. Kant), vonrohr@ipe.mavt.ethz.ch (P. Rudolf von Rohr).
*URL:* http://www.ltr.ethz.ch (M.A. Kant).

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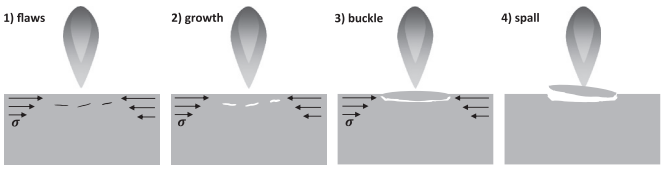



**Fig. 1.** Illustration of the spall formation on the rock surface, adapted from [7].

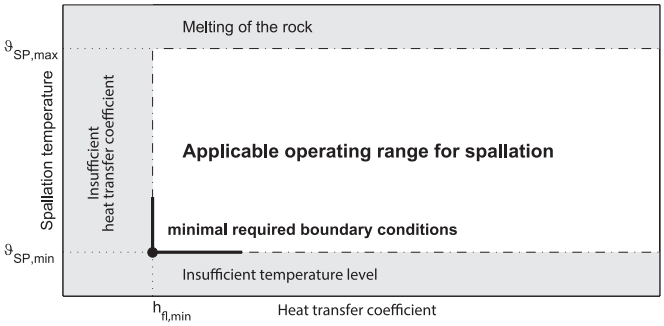



**Fig. 2.** Illustration of the boundary conditions limitations for the successful application of the spallation process.

able to resolve the spallation process. Therewith, an exact differentiation between spallation temperature $\vartheta_{SP}$ and temperature in the failure plane $\vartheta_{FP}$ is possible.

Apart from these experiments, different models based on theoretical and empirical approaches were developed to describe the effect of spallation and the appearing boundary conditions. Reviews about these models can be found in [2,15,10]. Thereby, the model developed by Rauenzahn [13], which uses Weibull's statistical theory of brittle failure, has to be particularly highlighted. Nevertheless, the spallation effect occurs at rates and scales and in rock environments that are difficult to experimentally probe.

Generally, by increasing the heat flux through the upper layer of the rock, the spallation frequency will increase and the thickness of the spalls will decrease.[13] On the other hand, the spallation temperature has to be kept under the melting point, which states the upper limit $\vartheta_{SP,max}$ of the boundary conditions range where spallation is applicable (see Fig. 2). Similar to this, a lower limit exists for the appearance of the spallation process. These minimal required boundary conditions are characterized by the lowest spallation temperature $\vartheta_{SP,min}$ and heat transfer coefficient $h_{fl,min}$ where spallation can be successfully initiated. Additionally, these conditions specify the size of the operating window where spallation can be utilized. In this report, the minimal required boundary conditions were defined as the conditions where at least a single spall could be detected by the measuring devices. Under this lower limit, the required heat transfer coefficients and surface temperatures are not reached and so the rock will heat up too slowly and no steep temperature gradient is created inside the rock, which will inhibit the spallation process, as only insufficient thermal stresses are created.

The optimal boundary conditions for spallation, where the highest spall performance can be achieved, will lie in between these two bounds.

Concluding, the minimal required boundary conditions define the lower limitation of the applicable operating range for spallation (see Fig. 2). Therewith, they can be used as boundary conditions for theoretical simulations of the spallation process and modeling approaches (e.g. failure mode determination, crack growth simulations, and grain scale failure analysis). Furthermore, they define the required operating conditions for spallation tools, as they specify appropriate flame temperatures and jet velocities. In order to optimize the application and to enhance the understanding of the process, both the upper and lower boundary condition limits have to be known. As the melting point is a property of the rock type and has been widely studied in the literature, this report focuses on the lower boundary condition limit, which is still poorly investigated.

To this aim, rock samples of Gotthard Granite (Gurtnellen, Switzerland) and Bethel Granite (Bethel, Vermont, USA) were rapidly heated by a methane–air burner. The spallation temperature was measured using high-speed pyrometers. With this setup the disintegration of single spalls could be monitored as well as the difference between spallation temperature $\vartheta_{SP}$ and temperature in the failure plane $\vartheta_{FP}$. As the pyrometer can temporally resolve the spallation process, the proposed measuring concept could contribute to the fundamental understanding of the spallation process. The heat transfer coefficient $h_{fl}$ (W/(m$^2$ K)) was determined by measuring the heat flux $q_{sens}$ (W/m$^2$) in the stagnation point of the impinging flame jet with an industrial heat flux sensor along with the fluid temperature $\vartheta_f$ and the surface temperature of the sensor $\vartheta_{s,sens}$ both in °C:

$$h_{fl} = \frac{q_{sens}}{\vartheta_f - \vartheta_{s,sens}} \quad (1)$$

## 2. Material and methods

In order to determine the boundary conditions during the spallation process, adequate measuring devices and an appropriate setup is required. Therefore, two different experimental setups for conducting the spallation experiments (determination of $\vartheta_{FP}$ and $\vartheta_{SP}$) and for measuring the heat transfer coefficient $h_{fl}$ are introduced in the following section. Additionally, in order to enhance the reliability of the reported values an error estimation was conducted.

### 2.1. Spallation experiments and infrared surface temperature measurements

#### 2.1.1. Experimental setup

The required flame jet to induce spallation is supplied by a 10 kW methane–air burner (Pharos E-354, see Fig. 3). The air and methane flow is controlled by mass flow controllers (Bronkhorst

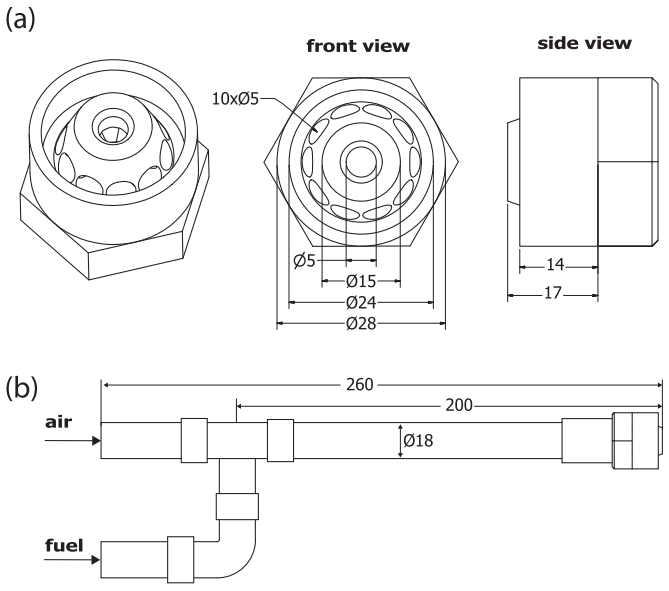



**Fig. 3.** (a) Schematic drawing of the nozzle with the most important dimensions. (b) Illustration of the burner assembly; dimensions in mm.

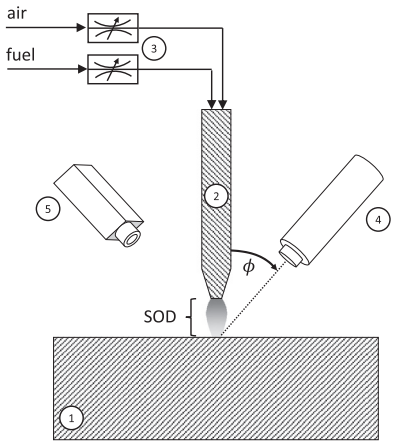



**Fig. 4.** Experimental setup for surface temperature measurements during the spallation process.

**Table 1**
Conducted experiments for the different rock types.

| SOD (mm) | Gotthard Granite | Bethel Granite |
|---|---|---|
| 50 | √ | √ |
| 80 | √ | √ |
| 110 | √ | √ |
| 130 | √ | √ |
| 140 | X | √ |
| 150 | X | √ |
| 160 | X | X |

EL-FLOW Select Series) and flame temperatures of about $\vartheta_f = 1400\,°C$ are created. The flame of the burner impinges orthogonally onto the 250 mm × 250 mm × 150 mm large, air dried rock samples.

The progression of the surface temperature during the spallation process is measured with two Optris OPTCTL3MH2CF2 pyrometers, which can measure temperatures from 200 °C up to 1500 °C with a sampling rate of 1000 Hz at a wavelength of $\lambda = 2.3\,\mu m$. The focus spot of the pyrometers has a diameter of 1.5 mm. With this high sampling rate and the small measuring spot it is possible to resolve the detachment of single spalls (see Section 3.1). Additionally, a video camera is installed, allowing a later visual observation of the experiments.

An illustration of the experimental setup can be found in Fig. 4, where (1) is the rock sample, (2) the burner with the adjusted standoff distance (SOD) between nozzle and rock surface, (3) the flow controllers for the air and the fuel flow, (4) one of the pyrometer and (5) the camera. In order to reduce disruptions from the surroundings the complete experimental setup is placed inside a metal box. The flame temperature was measured by using a thermocouple in the flow (see Section 2.2.1).

The minimal required boundary conditions were derived by adjusting constant burner flows ( $\dot{V}_{CH4} = 0.3\,N\,m^3/h$, $\dot{V}_{air} = 2.7\,N\,m^3/h$, based on the normal conditions of 0 °C and 1013.25 hPa(a)) and increasing the SOD in between the different conducted experiments (see Table 1). During the single spallation experiments the SOD was kept constant.

In the experiments marked with √ (see Table 1) spallation could be observed. In contrary, experiments marked with *X* no spallation could be detected anymore. The table shows that spallation will stop for higher SODs, whereupon differences in the rock types can already be observed, as Bethel spalled for higher SODs than Gotthard granite. For the further investigation of the boundary conditions, the evaluation of the appearing boundary conditions was limited to the most interesting SODs.

### 2.1.2. Rock material

Two granitic rock types were used in this study, commercially named as Gotthard granite (G) from Gurtnellen, Switzerland and Bethel White granite (B) from Bethel, Vermont, USA. Thin section photomicrographs of these rocks are shown in Fig. 5: Both rocks consist mainly of quartz, K-feldspar and plagioclase with minor amounts of biotite and muscovite. Both samples have variable grain sizes ranging with feldspar up to ca. 4 mm.

A summary of the most relevant thermophysical properties of the two rock samples can be found in Table 2. Thereby the thermal conductivities were determined by using the Kemtherm QTM-D3 equipment, the heat capacity was determined with a calorimetric measurement, Young's modulus and the unconfined compressive strength were determined with a compressive testing machine. All measurements were conducted at ambient temperatures. The values are in accordance with other literature data of granitic rocks.[16]

The thermophysical properties of quartz, the endmembers of the feldspar (anorthite, albite, and microcline) and phlogopite

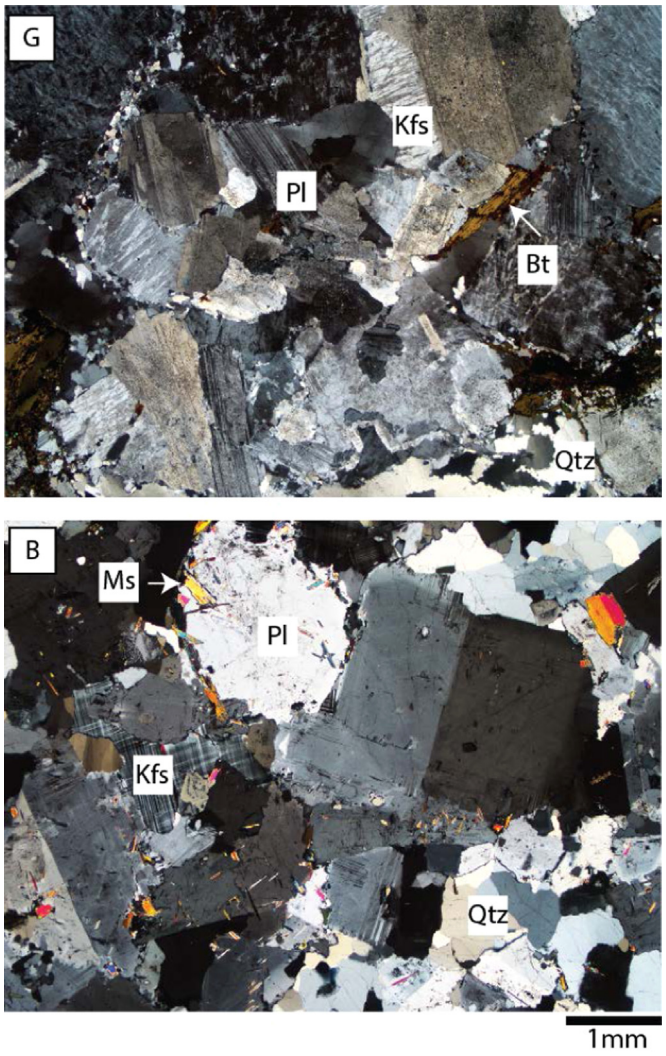



**Fig. 5.** Photomicrographs of the Gotthard (G) and Bethel (B) granite sample used in this study under cross polarized light. The rocks are mainly composed of quartz (Qtz), K-feldspar (Kfs), plagioclase (Pl) and minior amounts of biotite (Bt) and muscovite (Ms).

**Table 2**
Thermophysical properties of the used rock types, where $\lambda$ is the thermal conductivity, $c_p$ the heat capacity, $\rho$ the bulk density, $\kappa$ the resulting thermal diffusivity, $E$ Young's modulus and $UCS$ the unconfined compressive strength.

| Parameter | Bethel granite | Gotthard granite |
|---|---|---|
| $\lambda$ (W/(m K)) | 2.62 | 3.27 |
| $c_p$ (J/(kg K)) | 784 | 792 |
| $\rho$ (kg/m$^3$) | 2566 | 2594 |
| $\kappa$ (m$^2$/s) | $1.3\cdot10^{-6}$ | $1.6\cdot10^{-6}$ |
| $E$ (GPa) | 58 | 54 |
| $UCS$ (MPa) | 152 | 214 |

**Table 3**
Thermopysical properties of the minerals and the endmembers of the minerals present in the investigated rock types for two different temperatures ($p$=1 bar, data from FRENDLY [17], thermal conductivity values from [18]), $\alpha$ thermal expansion coefficient, $c_p$ – heat capacity, $\rho$ – density, $\lambda$ – thermal conductivity, $\kappa$ – resulting thermal diffusivity.

| Parameter | Temp. | Quartz | Anorthite | Albite | Microcline | Phlogopite |
|---|---|---|---|---|---|---|
| $\alpha\cdot10^{-6}$ (1/K) | 25 °C | 16.2 | 10.2 | 20.37 | 14.1 | 24 |
| | 600 °C | 4.29 | 16.2 | 50.4 | 21.9 | 37.6 |
| $c_p$ (J/(kg K)) | 25 °C | 747 | 761 | 791 | 734 | 850 |
| | 600 °C | 1132 | 1117 | 1261 | 1093 | 1232 |
| $\rho$ (kg/m$^3$) | 25 °C | 2624 | 2756 | 2616 | 2554 | 2788 |
| | 600 °C | 2536 | 2743 | 2564 | 2527 | 2736 |
| $\lambda$ (W/(m K)) | 25 °C | 7.7 | 1.7 | 2.3 | 2.3 | 2.2 |
| $\kappa\cdot10^{-6}$ (m$^2$/s) | 25 °C | 3.9 | 0.81 | 1.1 | 1.23 | 0.93 |

were calculated using the petrologic thermodynamic calculator FRENDLY from the Perple_X software package, by using the default thermodynamic database [17] and are summarized in Table 3.

Regarding Table 3, it can be seen that especially heat capacity and thermal expansion of the minerals change significantly with temperature. Due to the different thermal expansions of the minerals high thermal stresses are created inside the rock structure, leading to the local fracturing process described in this work. Furthermore, as the minerals do not heat up homogenously, additional thermal stresses are created.

### 2.1.3. Emissivity measurements

The emissivity of the investigated rock samples is a crucial point for determining the surface temperature during the spallation process. As the emissivity depends highly on the spectral range of the pyrometer, generally for an accurate measurement no literature data can be used and the emissivity has to be measured with the used pyrometer.[19] Additionally, the emissivity varies with the applied measuring angle $\phi$ (see Fig. 4). Therefore, the angle $\phi = 30°$ is kept constant for all described measurements.

Different ways to determine emissivities in a required wavelength range are reported in the literature. For the presented experiments, the determination based on a reference emissivity (Pyromark 2500 [20]) was applied. The procedure was performed corresponding to the reference emissivity material method described in ISO 18434-1:2008.[21] A part of a rock probe (size ca. $50 \times 50 \times 10$ mm) is coated with the reference paint creating the surface-modifying material. Pyrometer 1 (see Fig. 6) is focused on the surface-modifying material and Pyrometer 2 on the rock surface directly adjacent to the focus point of Pyrometer 1 (the focus points of the pyrometers are indicated by lasers). The rock probe is now heated up from underneath to the designated temperature by an electrical heater. As the emissivity of the surface-modifying material is known, Pyrometer 1 indicates the correct temperature. The emissivity setting of Pyrometer 2 is now adjusted until the displayed temperature is the same as the temperature indicated by Pyrometer 1. The final emissivity setting is the emissivity value for the adjusted temperature. The procedure is repeated for different surface temperatures of the rock and at least three times for every selected temperature. Thereby, enough time for the temperature to stabilize was allowed and for every measurement a new rock sample was used.

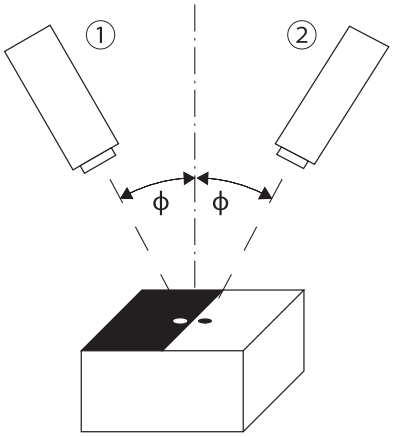



**Fig. 6.** Experimental setup for determining the emissivity of the investigated rock types at different surface temperatures.

The emissivity of the rock types were measured in the temperature range of $\vartheta_s$ = 400–675 °C and resulted in emissivity values of $\varepsilon_{Gotthard} = 0.549 \pm 12\%$ and $\varepsilon_{Bethel} = 0.434 \pm 17\%$.

It has to be considered that the reported values only account for the operating wavelength ($\lambda = 2.3$ μm) and the applied angle ($\phi = 30°$, see Fig. 4) of the installed pyrometers, as the emissivity values can vary significantly with different operating wavelengths and operating angles.

### 2.1.4. Error estimation for the surface temperature measurement

As spallation is a high-speed transient process, a reliable measurement procedure is necessary to accurately determine the surface temperatures during the process. Therefore, the influence of the uncertainty of the emissivity measurement on the surface temperature measurement was investigated and the error induced by the impinging flame was identified. Additionally, the accuracy of the pyrometer is considered, which is given by the manufacturer of the pyrometer and can be calculated according to the following equation:

$$\Delta\vartheta_{sys} = \pm (0.3\cdot10^{-3}\cdot\vartheta_{obj} + 2\ °C) \quad (2)$$

where $\Delta\vartheta_{sys}$ is the systematic error of the devices in °C and $\vartheta_{obj}$ the surface temperature of the object in °C.

The influence of the uncertainty of the emissivity was determined by changing the emissivity setting of the pyrometer to a lower or higher value than the correct one and monitoring the received temperature difference of the pyrometer at a reference surface temperature of $\vartheta_{ref}$ = 610 °C. This procedure was also done for other reference surface temperatures within the range of application, showing similar results. The collected data is summarized in Fig. 7.

Fig. 7 shows that uncertainties in the emissivity measurement have a relatively low impact on the surface temperature measurement.

At low surface temperatures (<400 °C) the impinging flame is considerably influencing the measurement. If the surface temperature increases, this influence is significantly dropping. This additional error source was investigated and considered in the overall error estimation.

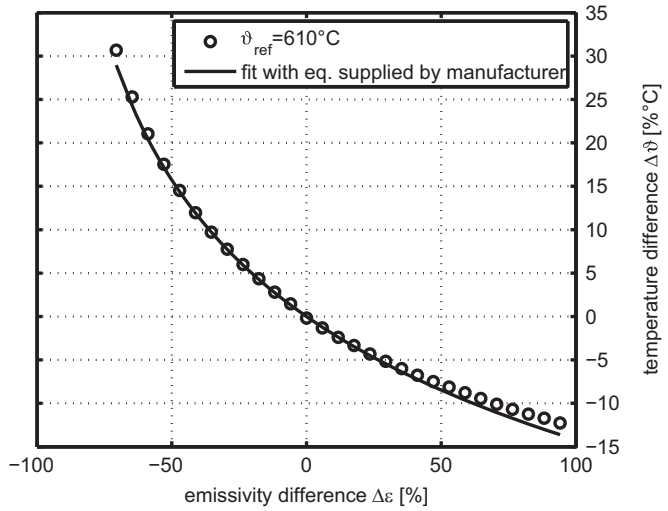



**Fig. 7.** Sensitivity of the emissivity uncertainty on the temperature measurement.

Combining the error due to the uncertainty of the emissivity measurement, the influence of the flame on the surface temperatures measurement and the accuracy of the pyrometers, the total uncertainty of the surface temperature measurement can be derived (Fig. 8). As the relative uncertainty accounts for only 5–6%, in the temperature range interesting for spallation, accurate surface temperature measurements are possible with the described measuring principle. Additionally, the absolute uncertainty function presents a minimum at a surface temperature of approx. $\vartheta_{SP} = 500\,^{\circ}C$, which is near the temperature where spallation occurs, as shown in Section 3.3. In the presented total uncertainty calculation, the highest uncertainty from the emissivity measurement of all rock types was considered.

### 2.1.5. Processing of the raw data

Fig. 9(a) shows a typical surface temperature over time plot of the experimental raw data. It can be observed that the temperature rises within a few seconds to about 600 °C and spallation starts, indicated by a rapid temperature drop and followed by a next heating period until the next spall detaches. Additionally, spalls can significantly overheat, if they are not removed by the momentum of the impinging flame jet and spallation temperatures of about $\vartheta_{SP} = 1000\,^{\circ}C$ can be reached (see Fig. 9(a) at $t \approx 90$ s). When the spall finally detaches from the surface, the temperature drops back to the surface temperature before the spall overheated.

In Fig. 9(b) it can be seen that the applied measuring principle is able to resolve the detachment of a single spall. This ability was defined as a crucial parameter to improve the research on the spallation process [14] and became recently possible with the availability of high-speed pyrometers. Fig. 9(b) shows that the spallation process induces a temperature difference of approx.

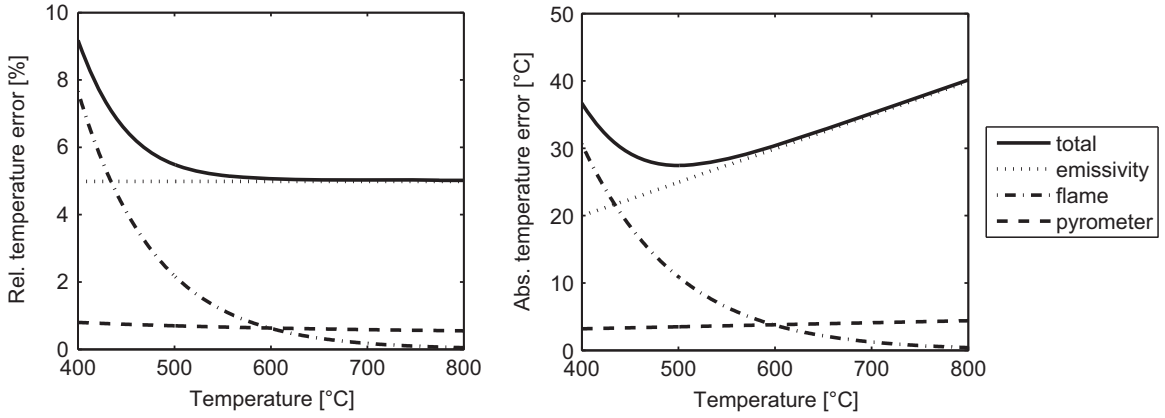



**Fig. 8.** Relative and absolute uncertainty for the temperature measurement during the spallation experiments; total: total error; emissivity: error due to the uncertainty in the emissivity measurements; flame: error due to the influence of the flame; pyrometer: accuracy of the pyrometer.

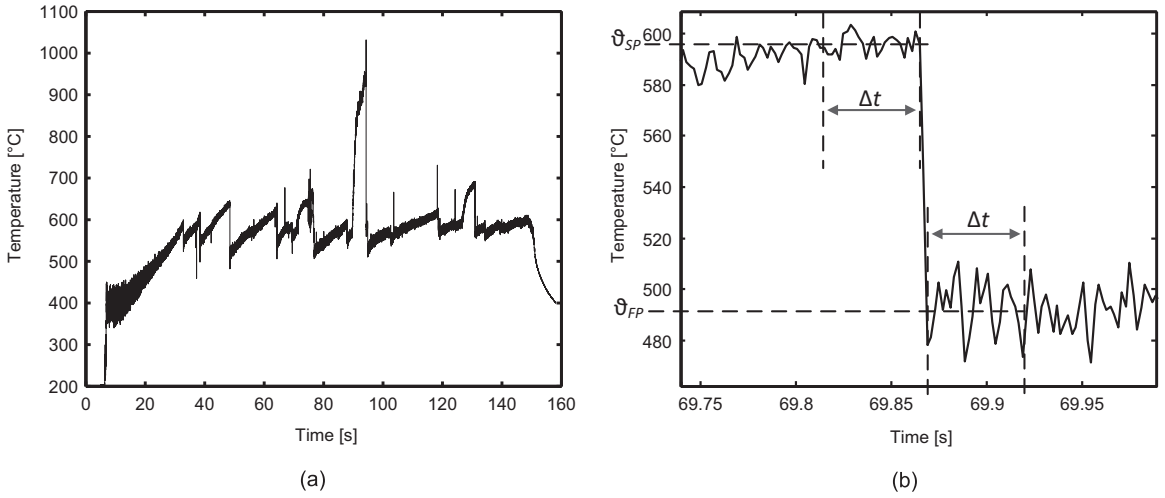



**Fig. 9.** (a) Typical figure of the acquired raw data of the temperature measurement during spallation. (b) Detachment of a single spall.

$\Delta\vartheta = 100\,^{\circ}C$ in a few milliseconds. In order to exactly determine the temperature before and after the detachment of a spall, the temperature values are averaged over a time period of $\Delta t = 50$ ms. Therewith, the influence of the induced fluctuations by the flame is minimized.

### 2.2. Stagnation point heat transfer coefficient measurements

As described above, the heat transfer coefficient in the stagnation point of the impinging flame jet could not be determined during the spallation process, as no direct heat flux measurement on the rock was possible. Therefore, an additional setup is required where the rock is replaced by a heat flux sensor.

#### 2.2.1. Setup for stagnation point heat transfer coefficient measurements

In order to determine the heat transfer coefficient, the same experimental setup as described in Section 2.1.1 is used, where the rock sample is replaced by a sensor plate, with the same dimensions as the rock sample. The sensor plate consists of a heat flux sensor (Wuntronic FCR High-Temperature Heat Flux Sensor) integrated into a housing, which is supplied with a water cooling circuit, in order to regulate the surface temperature of the sensor.

During the operation, the sensor generates a voltage $U$ which is directly proportional to the heat flux $q_{sens} = U/S$, where $S$ is the sensitivity of the sensor. This voltage is recorded for the different conducted experiments (see Table 1) along with the surface temperature of the sensor $\vartheta_{s,sens}$ and the fluid temperature $\vartheta_f$ of the flame.

The surface temperature of the sensor $\vartheta_{s,sens}$ was determined using the same pyrometer as described in Section 2.1.1 and by coating the sensor also with the reference paint. The flame temperature $\vartheta_f$ was measured 10 mm above the stagnation point using K-type thermocouples. As conduction and radiation can play a significant role at these high temperature ranges, the flame temperature was corrected using the method suggested by Pitts et al. [22]. In order to apply the suggested method, the flame temperature was measured using thermocouples with three different diameters ($d$=0.5 mm, 1.0 mm, 1.5 mm) and the results were prolongated for a thermocouple with zero diameter. With heat flux $q_{sens}$, surface temperature $\vartheta_{s,sens}$ and fluid temperature $\vartheta_f$ known, the heat transfer coefficient of the impinging flame jet $h_{fl}$ can be calculated using the following equation:

$$h_{fl} = \frac{q_{sens}}{(\vartheta_f - \vartheta_{s,sens})} \tag{3}$$

#### 2.2.2. Convective calibration of the applied heat flux sensor and determination of the heat flux through the rock

As the applied sensor is only calibrated for low temperature ranges, an additional convective calibration was accomplished using a convection calibration setup [10,23,24] and a reference sensor (Vatell HFM-8E/H). The reference-sensor has been calibrated by the manufacturer using instruments which accuracy is traceable to NIST. The calibration of the applied sensor is accomplished by measuring the sensitivity $S$ of the sensor for different surface temperatures $\vartheta_{s,sens}$. Thereby, the calibration procedure suggested by Meier et al. [24] was applied. The calibration of the sensor resulted in a constant sensitivity of $S = 0.404\ \mu V/(W/m^{-2}) \pm 15\%$ for surface temperatures of the sensor in the range of $\vartheta_{s,sens} = 225$–$460\,^{\circ}C$. As no increasing or decreasing trend was visible over the investigated temperature range, the same sensitivity value was used for temperatures exceeding 460 °C.

With the sensitivity of the heat flux sensor $q_{sens}$ known, the heat transfer coefficient during the flame impingement studies $h_{fl}$ can be calculated by the following equation:

$$h_{fl} = \frac{q_{sens}}{(\vartheta_f - \vartheta_{s,sens})} = \frac{U}{S(\vartheta_f - \vartheta_{s,sens})} \tag{4}$$

The heat flux through the rock samples can be finally calculated by assuming that the heat transfer coefficients are identical for the spallation experiments and the flame impingement studies. This assumption is valid, if the fluid-side conditions and the thermophysical properties of the impinging fluid in the boundary layer are similar. Thereby, the thermophysical properties in the boundary layer mainly depend on the surface temperature of the impinged object.

As the geometry of sensor plate and rock are similar and the same flow rates and SODs were adjusted, the fluid-side conditions of the two experiments can be considered as comparable. In order to avoid high differences in the surface temperatures between rock and sensor, the used sensor is equipped with a low conductivity steel surface. Therewith, surface temperatures of $\vartheta_{s,sens} = 390$–$550\,^{\circ}C$ (depending on the adjusted SOD, compare Section 3.2) could be achieved. As the difference in the surface temperatures between spallation experiments and flame impingement studies was at its maximum $\Delta\vartheta_{S,R} = 180$ K, it can be concluded that the physical properties of the fluid do not change significantly. Therewith, the assumption that the heat transfer coefficient is similar is valid and Eq. (5) can be used to calculate the heat flux through the rock. Thereby, $\overline{\vartheta}_{SP}$ is the mean spallation temperature averaged over all spalls detected for the individual experiment:

$$q_{rock} = h_{fl}(\vartheta_f - \overline{\vartheta}_{SP}) \tag{5}$$

In order to precise the measurements, an error estimation was carried out for the heat transfer coefficient measurements: the error due to the calibration of the sensor was investigated and the uncertainty of the fluid temperature measurement was estimated. The error of the surface temperature measurement was already introduced in Section 2.1.4. The error estimation showed that the uncertainty in the calibration of the heat flux sensor dominates the error estimation and the other uncertainties only play a minor role. With the proposed measuring principle heat transfer coefficients can be measured with an uncertainty of about 17%.

## 3. Results and discussion

### 3.1. Spallation temperature measurements

As stated above, two different rocks were investigated in this report: Gotthard Granite (Switzerland) and Bethel Granite (USA). For every experiment the post-processing as described in Section 2.1.5 was applied. The experimental setup introduced in Section 2.1.1 was used and for every investigated SOD and rock type three different experiments with different rock samples were conducted.

Fig. 10 shows a representative plot for the spallation temperatures $\vartheta_{SP}$ and the temperatures in the failure plane $\vartheta_{FP}$ for a single spallation experiment conducted on Gotthard Granite. Thereby, on the $x$-axis the chronological sequence of the detected spalls is plotted.

It can be seen that the spallation temperature $\vartheta_{SP}$ fluctuates highly within the different detected spalls. This is a result due to the fact that the experiments were conducted at the minimal required boundary conditions. Therewith, the appearing conditions are just enough for a successful spall formation and the created spalls vary considerably in their geometry. On the other hand, the fluctuation in the temperature in the failure plane of the spall $\vartheta_{FP}$ are significantly smaller. Further investigations of this behavior will be done in the future.

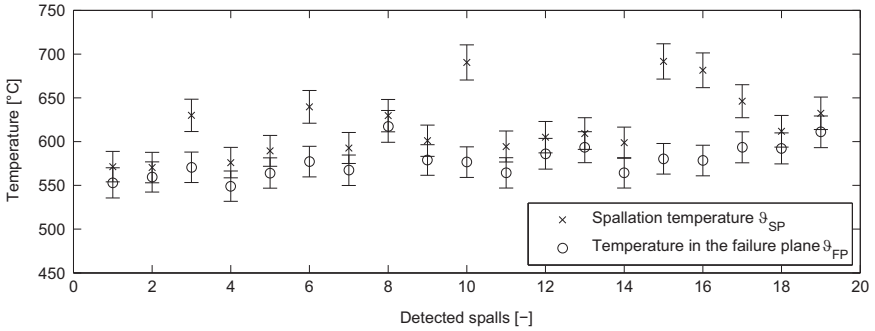



**Fig. 10.** Representative results of surface temperature and temperature in the failure plane for Gotthard granite.

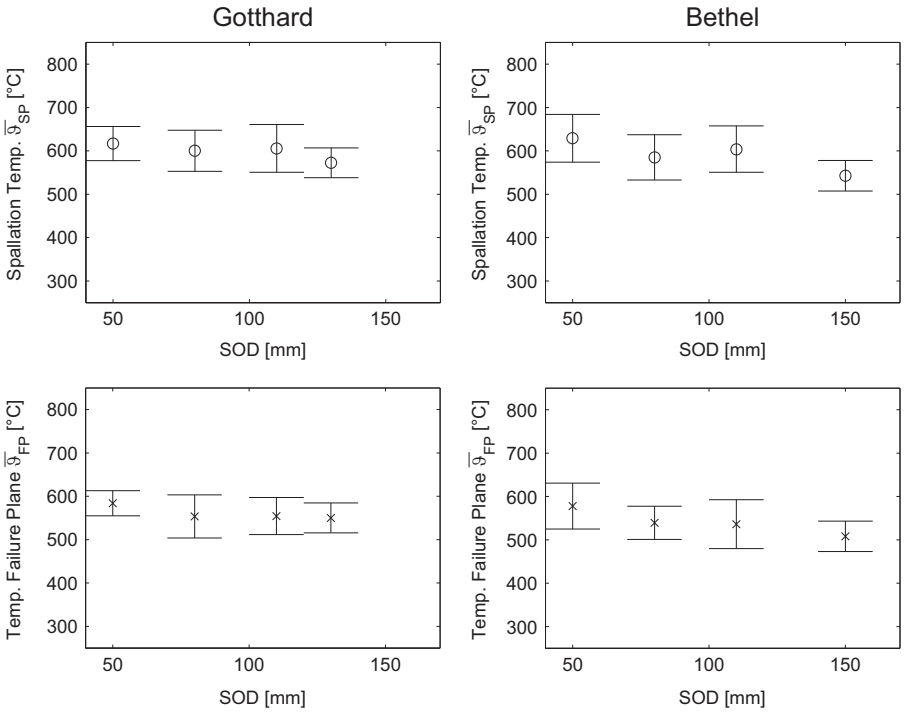



**Fig. 11.** Summary of the observed mean spallation temperatures and mean temperatures in the failure plane at different SODs for Gotthard and Bethel Granite.

Fig. 11 shows the mean spallation temperatures $\overline{\vartheta}_{SP}$ and mean temperature in the failure plane $\overline{\vartheta}_{FP}$ averaged over all spalls detected for the individual experiments. Thereby, the temperature range indication does not show the measurement error, but indicates the deviation in-between all detected spalls. This is significantly higher than the measuring error which accounts for only 5–6% of the spallation temperature as described in Section 2.1.4.

The mean spallation temperature for all conducted experiments is around $\overline{\vartheta}_{SP} = 600\,°C$, whereas the mean temperature in the failure plane is lower at around $\overline{\vartheta}_{FP} = 550\,°C$. Thereby, it has to be considered that these temperatures are averaged values. The temperature difference between spallation temperature $\vartheta_{SP}$ and temperature in the failure plane $\vartheta_{FP}$ for a single spall can be considerably higher or lower. Additionally, it can be seen that these two temperatures vary considerably among all conducted experiments.

### 3.2. Stagnation point heat transfer coefficient measurements

Fig. 12 and Table 4 show the determined heat transfer coefficient $h_{fl}$ for the different experiments (see Section 2.1.1). The heat flux $q_{sens}$ was measured with the calibrated heat flux sensor (see Section 2.2.2) in the stagnation point of the impinging flame jet.

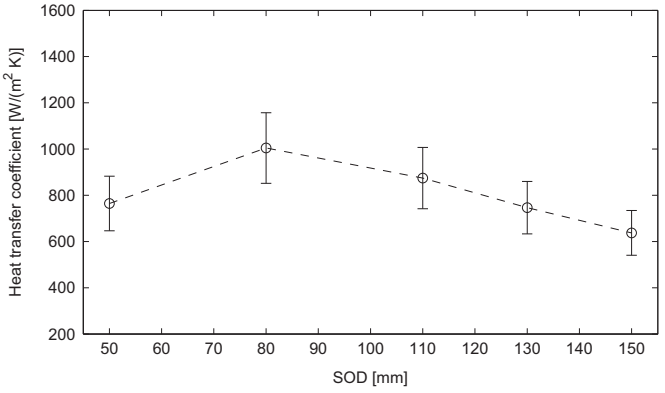



**Fig. 12.** Results of the heat transfer coefficient in the stagnation point of the flame jet.

The stagnation point indicates the center point of the impingement where the middle stream line hits the impingement plate and the local velocity is zero. This point was determined by introducing a probe into the nozzle of the burner, indicating the center point. Additionally, before every measurement, the sensor was moved in a plane orthogonally to the burner to assure that the heat flux

**Table 4**
Summary of the measured data for different SODs; $q_{sens}$ – heat flux measured with the sensor, $\vartheta_{s,sens}$ – surface temperature of the sensor, $\vartheta_f$ – flame temperature, $h_{fl}$ – resulting heat transfer coefficient.

| SOD (mm) | $q_{sens}$ (kW/m$^2$) | $\vartheta_{s,sens}$ (°C) | $\vartheta_f$ (°C) | $h_{fl}$ (W/(m$^2$ K)) |
|---|---|---|---|---|
| 50 | 720 | 511 | 1450 | 766 |
| 80 | 866 | 552 | 1410 | 1009 |
| 110 | 755 | 495 | 1360 | 873 |
| 130 | 609 | 435 | 1250 | 747 |
| 150 | 450 | 389 | 1100 | 633 |

profile is symmetrical around the selected stagnation point. The fluid temperature $\vartheta_f$ and the surface temperature $\vartheta_s$ were determined by following the principles reported in Section 2.2.1.

According to Fig. 12, a maximal heat transfer coefficient in the stagnation point of around $h_{fl} = 1000$ W/(m$^2$ K) at a SOD of 80 mm was reached during the impingement experiments. Thereafter, the heat transfer coefficient declines for increasing and decreasing SODs. This characteristic maximum at a specific SOD was already observed in other investigations using impinging isothermal gas and flame jets.[25]

### 3.3. Minimal required boundary conditions

If the reported mean spallation temperatures $\overline{\vartheta}_{SP}$ and the corresponding stagnation point heat transfer coefficients $h_{fl}$ respectively heat fluxes $q_{rock}$ are combined, the minimal required boundary conditions for the successful initiation of the spallation process can be found (see Fig. 13).

Fig. 13 shows the different measured boundary conditions for the investigated experiments. The boundary conditions with the lowest temperature respectively heat transfer coefficient can be regarded as the minimal required boundary conditions. If the required heat transfer coefficients are not reached, the rock will heat up too slowly and no steep temperature gradient is created inside the rock. Additionally, if the surface temperature is not above the stated limit, the temperature difference between initial rock temperature and surface temperature will be too low. Both will inhibit the spallation process as only insufficient thermal stresses are created.

According to the gained data, first spalls can be detected at spallation temperatures of around $\vartheta_{SP,min} = 500$ °C and heat transfer coefficients of around $h_{fl,min} = 500$ W/(m$^2$ K). Fig. 13 shows that Bethel granite starts at slightly lower temperatures than Gotthard granite, due to different physical properties of the two rock types. It can be concluded that the reported minimal required boundary conditions will change according to the rock types and rock samples, as the physical properties will vary considerably due to the natural composition of the rock. Nevertheless, the range of the boundary conditions where spallation can be first expected will remain the same.

With the stated minimal boundary conditions, the determination of the applicable operating range is possible. Therewith, these conditions contribute to investigations of the optimal process parameters and so they enhance a possible application of the spallation process for e.g. drilling purposes. Additionally, they are a basis for simulations of the spallation process (e.g. failure mode determination, crack growth simulations, and grain-scale failure analysis) and they define the required operating conditions for spallation tools, as appropriate flame temperatures and jet velocities are specified.

It must be highlighted that the reported boundary conditions state the minimum conditions at which spallation was possible. A successful, efficient application of this process must be conducted at significantly higher heat transfer coefficient.

### 3.4. Comparison with data available from literature

In order to confirm the reliability of the proposed measuring principle, the reported data is compared in Fig. 14 with data available in the literature.[13,14] These reports did not focus on the minimal required boundary condition, but as the same heating principle of the rock were used (impingement of a flame jet), the characteristic of the appearing boundary conditions should agree with the ones stated in this report. It has to be considered that the data reported by Wilkinson [14] and Rauenzahn [13] was gained using different rock types than used in this report.

Fig. 14 shows that the reported boundary conditions from this study are well within the range reported by Wilkinson. Both data show that the spallation temperature increases linearly with the

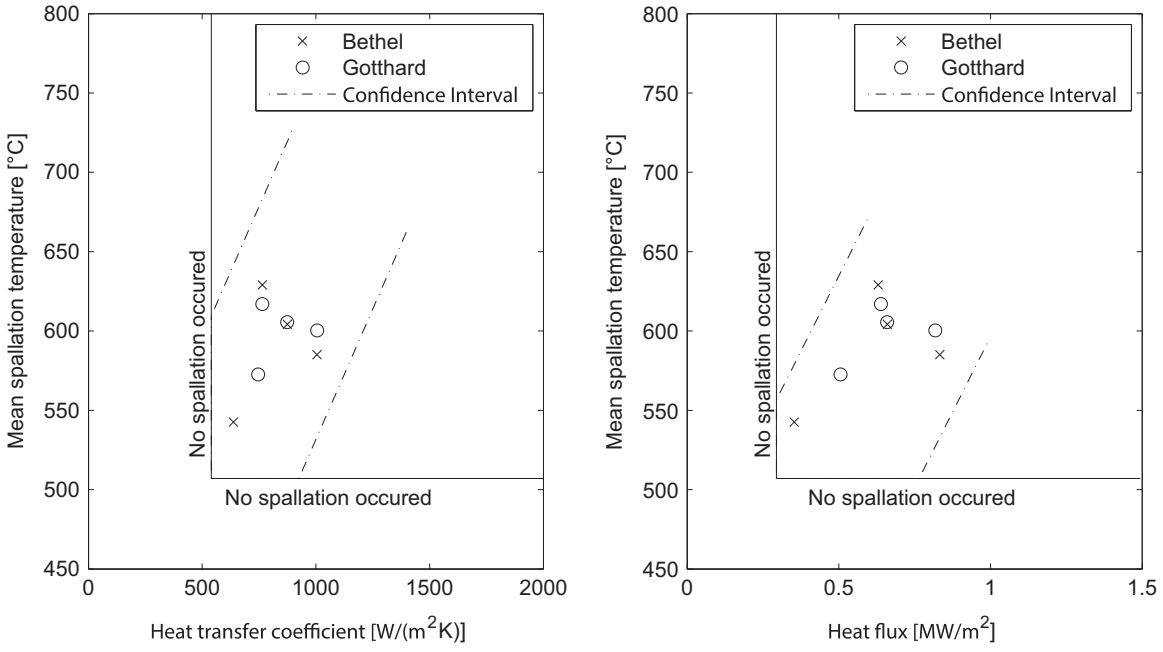



**Fig. 13.** Summary of the reported minimal required boundary conditions.

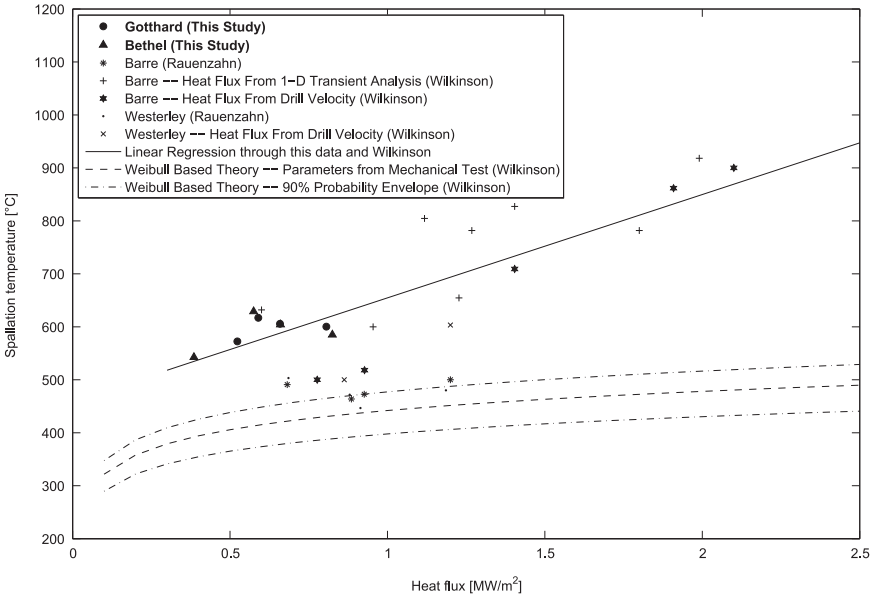



**Fig. 14.** Comparison of the measured spallation temperatures and heat fluxes with the data reported by Wilkinson and Rauenzahn, Weibull Data for Barre Granite from [14], assuming physical properties at 573 K (Figure adapted from [14]).

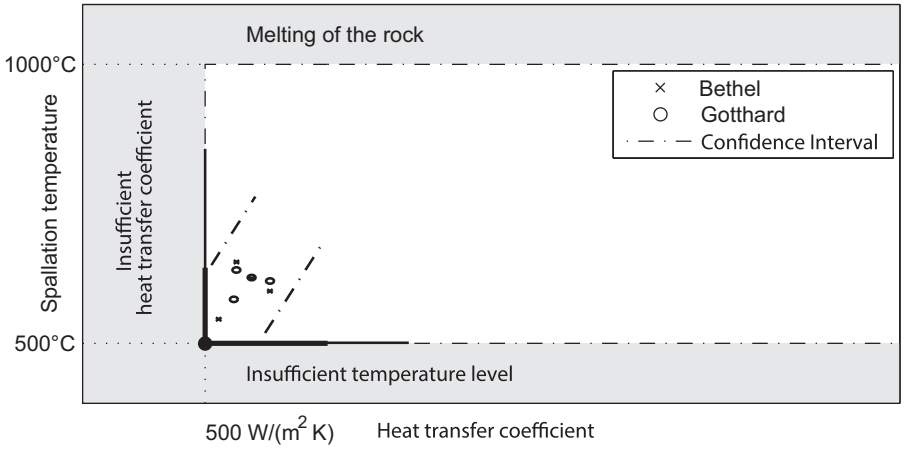



**Fig. 15.** Applicable operating range for thermal spallation drilling for the investigated rock types.

heat flux. Additionally, the literature values support the minimal required boundary condition derived in Section 3.3, as also no values are reported under the stated limitations.

The report supports Wilkinson's statement [14] that the model developed by Rauenzahn seems to underestimate the occurring spallation temperatures during the spallation process for this specific heating principle, if the required parameters are obtained from mechanical experiments. Therefore, an application of this model for spallation processes might not be possible and the adaption of the model or a development of a new model in accordance with the reported measurements might be required. It has to be highlighted that a fitting of the model to the received data is possible and shows good accordance with the experimental data.[14]

Nevertheless, the validation of Wilkinson's data for higher heat fluxes remains so far open and will be done in the future. Additionally, measurements with other heating principles should be pursued, as therewith the whole operating range can be investigated (see Fig. 2). Furthermore, according to the data described in this report and by Wilkinson, it can be concluded that the $\alpha$–$\beta$ quartz phase transition in granite, appearing at about 573 °C, and implying a significant increase of the thermal expansion,[26] could contribute to the spallation process. Nevertheless, the role it plays remains unknown and has to be investigated in further research.

## 4. Conclusions and outlook

In the present report a novel measuring concept for the determination of the boundary conditions during the spallation process was reported. The procedure is able to resolve the detachment of single spalls during the spallation process and to determine the temperature before and after the detachment, with an accuracy of 5–6% of the measured temperature. Therewith, it is a useful tool for further fundamental investigations of the spallation process. The heat transfer coefficient was estimated by measuring the heat flux in the stagnation point of the impinging flame jet using an industrial heat flux sensor. With this measuring principle two different granitic rock types were investigated for the minimal required boundary conditions which are required for a successful initiation of the spallation process. Spallation was observed for heat transfer coefficients higher than 500 W/(m$^2$ K) and for surface temperatures well above 500 °C.

With these minimal required boundary conditions and the melting point of the rocks, which is approximately 1000 °C for granitic rocks at atmospheric pressures,[16] the applicable operating range for thermal spallation of granitic rocks can be determined (see Fig. 15).

Concluding, the reported data shows good accordance with data published by Wilkinson [14] and supports the proposed linear characteristic of the applied heating principle. The measured minimal required boundary conditions define the limitations of the applicable operating range of the spallation process. Therefore, they can be used as boundary conditions for theoretical simulations of the spallation process and modeling approaches (e.g. failure mode determination, crack growth simulations, and grain-scale failure analysis). Furthermore, they define the required operating conditions for spallation tools, as they specify appropriate flame temperatures and jet velocities. In order to enhance the current understanding of the spallation effect and to facilitate the application of the process for e.g. drilling purposes or to prevent the appearance of spallation in tunnel fires, the whole operating range should be investigated. Therefore, further investigations with other heating principles and research on the optimal boundary conditions should be pursued. These investigations would benefit from a reliable method for measuring the heat transfer coefficient directly on the rock.

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## Appendix A. Supplementary data

Supplementary data associated with this paper can be found in the online version at http://dx.doi.org/10.1016/j.jfranklin.2015.02.023.

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